Walmart A/B Test Analysis

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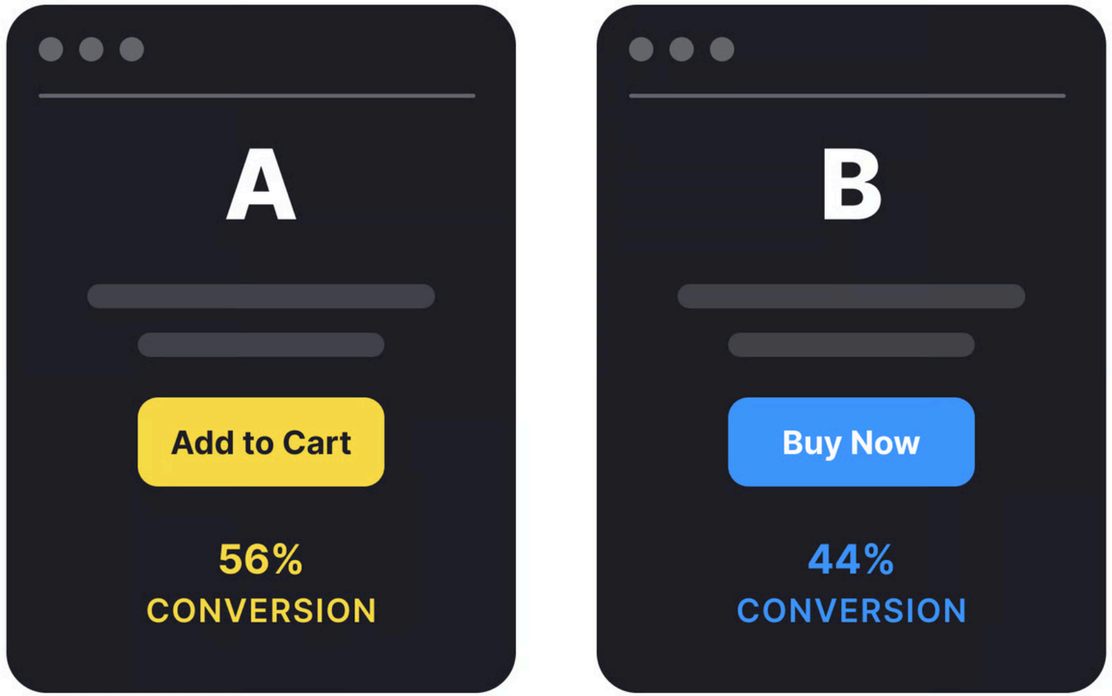
Introduction

In this case study we analyze an A/B test conducted by Walmart to evaluate a new product bundling recommendation at checkout.

The goal of the experiment is to understand whether the new feature improves customer behavior and overall revenue.



Conversion Rate Comparison

- The control group shows a conversion rate of 11.8%, while the treatment group has 10.6%.

- The difference between the two groups is relatively small (1.2 percentage points).

The t-test shows no statistically significant difference (p = 0.2317).

```
data:  conversion by treatment
t = 1.1961, df = 3780.6, p-value = 0.2317
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -0.007822122  0.032298254
sample estimates:
mean in group 0 mean in group 1
      0.1180223       0.1057843
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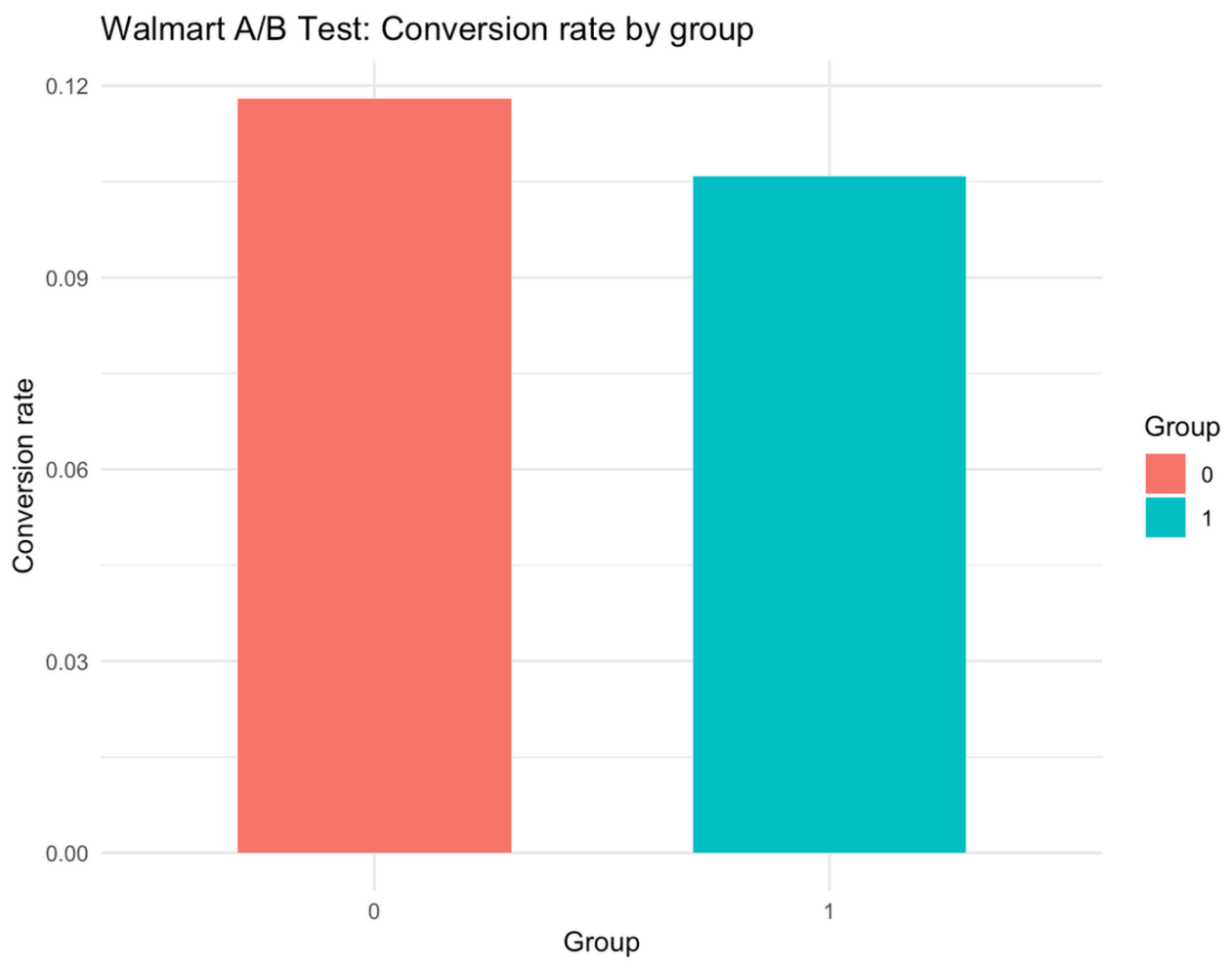
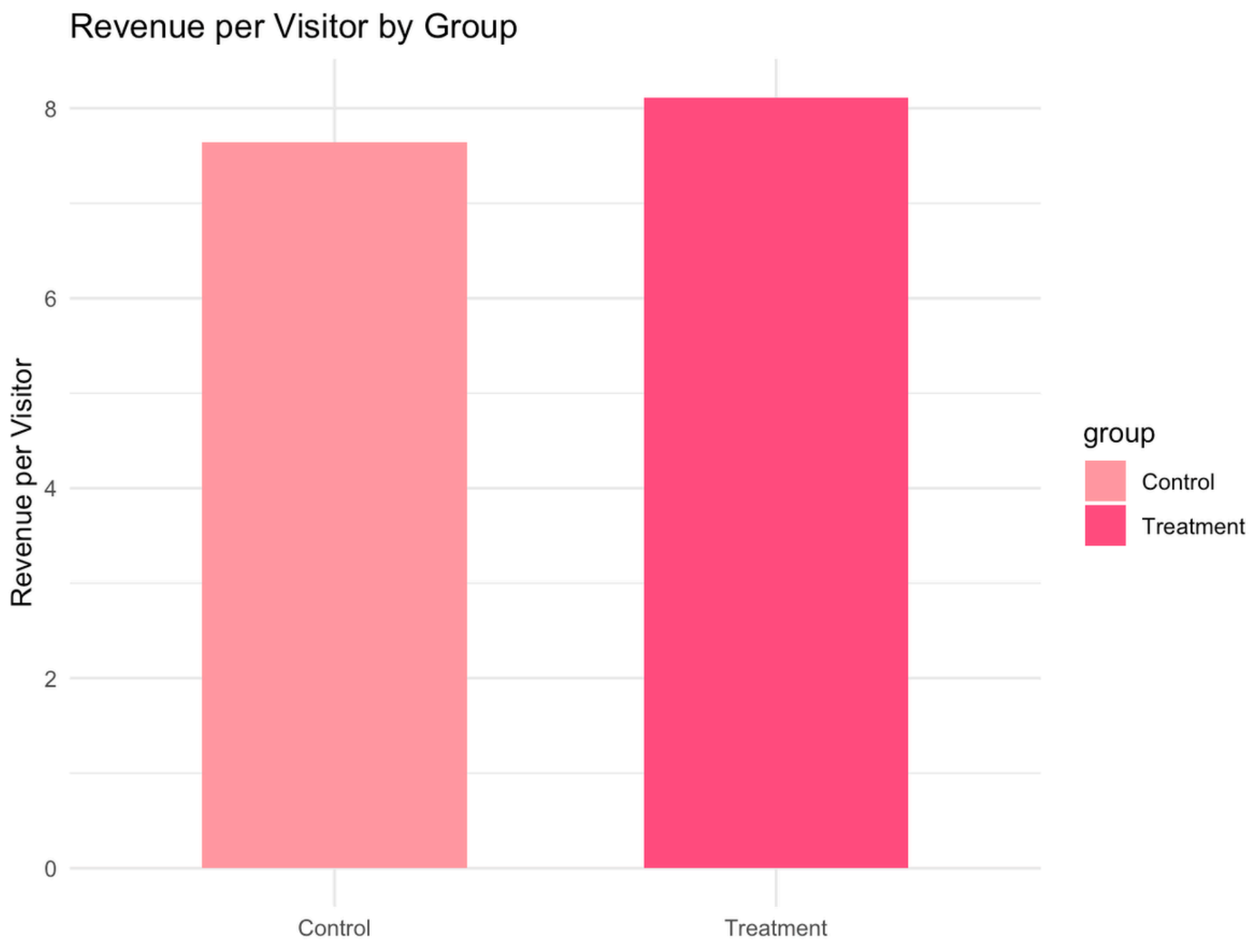


Revenue per Visitor

- The treatment group shows higher revenue per visitor (8.11) compared to the control group (7.64).

- This increase is driven by a higher average order value (76.7 vs 64.7).

- This suggests that the bundling recommendation may encourage customers to spend more per purchase.
- However, the difference in revenue per visitor is not statistically significant (p = 0.5707).



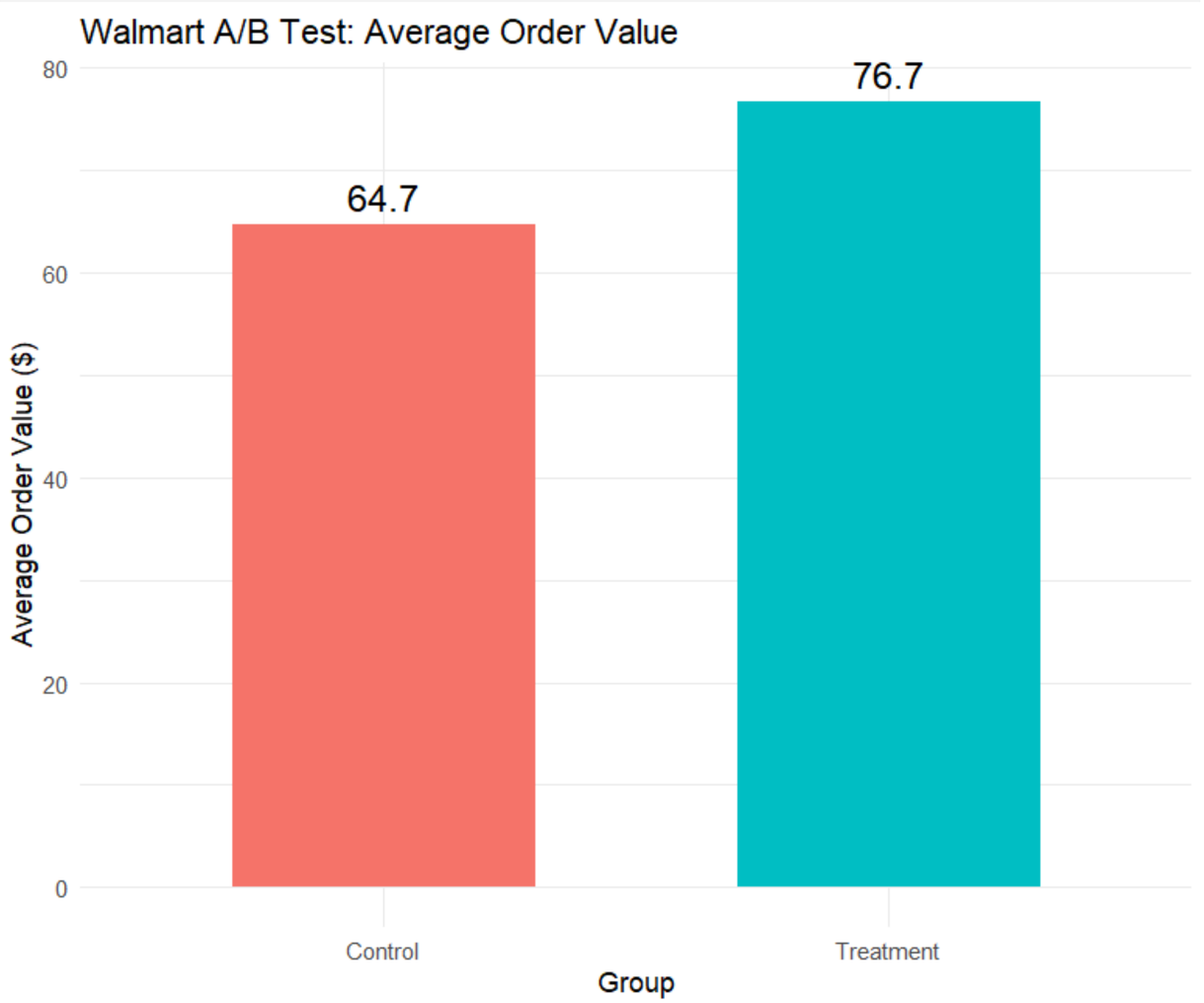
Revenue Impact of Product Bundling

Key Insight

- Conversion remains similar
- Average order value increases

Business Impact

- Customers buy more items per order
- Larger basket size
- Higher revenue per purchase



+18% Average Order Value
+ $12 per order

Which Metric Matters Most?

Key Metrics

- Conversion rate slightly decreases
- Average Order Value increases significantly
- Revenue per user increases

The most relevant metric for revenue growth

Revenue per user - captures both conversion and order value

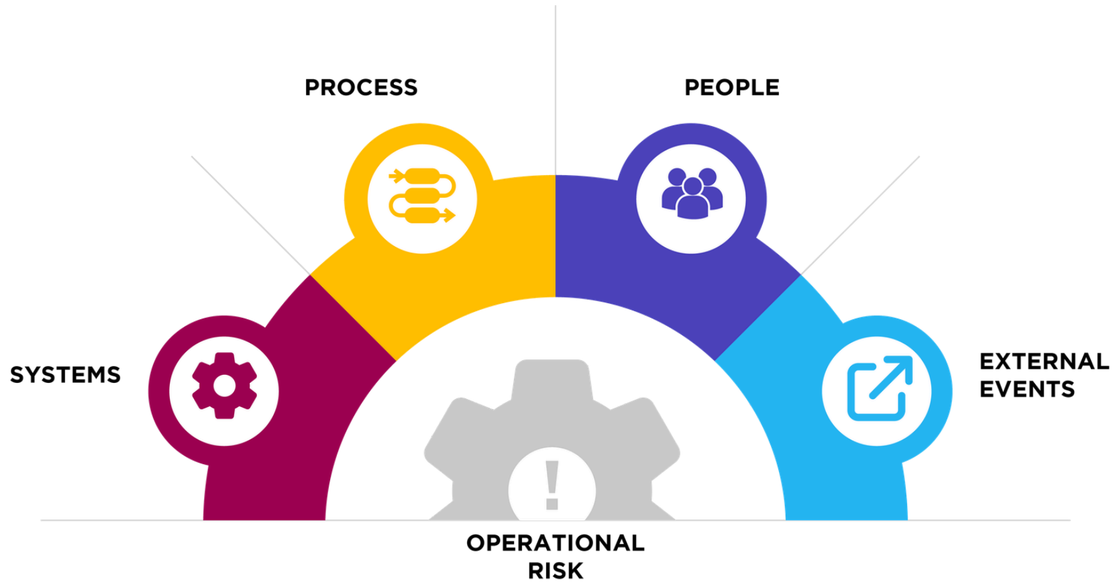
Metric	Control	Treatment
Conversion Rate	11.8%	10.6 %
Revenue per User	$ 7.64	$ 8.11
Average order value	$ 64.7	$ 76.7

Revenue per User ↑
Average Order Value ↑

What operational risks should Walmart check before rollout?

Before launching the bundling feature, Walmart should evaluate several operational risks:

- **Inventory risk** – bundled products must remain in stock.
- **Customer experience** – bundles should not confuse or overwhelm customers.
- **Pricing perception** – customers must still perceive bundles as valuable.
- **Checkout speed** – the recommendation system should not slow down the checkout process.
- **Margin impact** – discounts within bundles should not reduce profit margins too much.



Recommendation for the merchandising team

Based on the analysis, Walmart should continue testing the bundling recommendation, as it has the potential to increase revenue per order.

Although conversion did not significantly increase, the treatment shows higher average order value and revenue per user.

Therefore, Walmart should:

- **refine the bundle recommendations**
- **test different bundle combinations**
- **monitor revenue per visitor and profit margins**

If future tests confirm revenue growth, the feature should be rolled out more broadly across the platform.

THANK YOU!